

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA04 **COURSE CODE:** Operating Systems

COURSE OUTCOME COVERED

***C01:** Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 Marks:25

Code of Conduct:

Annexure A

SHORT ANSWER TEST

Total

DS. Kaubash

C19052485g).

(Name / Reg No) certify that this submission *is my original work* and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

S.

ЦЕВ а

Signature of the *student*:

1

I classify different types of

operating Systems

A modern, operating System
that Supports multiple users
and multiple processes
requires efficient.

The Suitable operating
resource
sharing.

System types are

Multi-user

operating System

1) Allow many users to
access the computer

semaltaneously

bal lion

in) Share CPU, memory files,
and devices among

ig

CBO,

ii) provides security and
user authentication. Eg: Linux,
Unix the appro

Aspard

rault tasking operating
System .

process

user

1) Executes multiple programs simultaneously
1) uses CPU scheduling to switch between

cpu

it is Improves CPU utilization and system Performance.

(a) **dad** med 100

Eg: windows, Linux.

parallel processing operating System.

1.

1) uses more than one `cpu/core`. 1.)

multiple processors execute
processes Simultaneously...?

""") increase speed and
reliability.

4. Time

Fin

Sharing operating

System

id

Cpu time is divided into
small tha

1) Every user gets
equal cpu tim """)

Provides fast response to
user operating System

Structure

USERS

ponto Application Programs
system call interface

Itt

operating System.....

Goivor (

Process management in

un't yo

Memory Management File
Management Device

cagnagement Security

o be

Hardware

اون

CCPU, RAM, DISK, Y0)

Sequence of System calls.

fork creates a child processes

exec() → Loads a new program

wait C → parent waits for child

exit child terminates. I

```
timi
```

```
if (pid == 0)
```

```
{
```

```
    execl ("/bin/ls", "ls", NULL);
```

```
}
```

```
else
```

```
{
```

```
    wait (NULL);
```

y

exit (o);

a) Categorize System calls used for file management

System calls.

Create()

open

ready

writer

close c delete c

rename ()

Functions:

create files

Read word te data

Delete file **Rename files.**

2004

Device Management system calls.

D

request device release device

read device

write device

Poctlly

Functions:

J

Allocate devices

ii) Read from **devices**

iii) write to devices

ردا

((AUM) tow

(C) be

Progoto) (5

1) Control hardware settings

Layared structure

Users

programs

1

System Call Interface

program

1

File System

1

Device Drivers

<

(

o saulo

ware Devices .

Hardwo

```
#include <stdio.h> # Indude <fcntl.h>
```

```
#include <unistd.h>
```

```
int main
```

```
{
```

```
int fd;
```

```
~^  
c
```

```
alit
```

3

```
fd = open ("example.txt",  
O_RDONLY) ;  
read Cfd, buffer, 100) ;  
close (fd);  
return 0;
```

RSAU

bung 2018

a) (Identify the type of operating System

The Suitable operating System is a Real Time operating System (RTOS).

- Produces immediate response. Meets strict deadlines"

ii)

(ii) High reliability thus robustly used in embedded System

Eg

~) predictable exceaustion

abo

Nx Worke

PNX

swer boxenda cobar

Free PTOS

bool

atwolor bok only

Applications.

CRI). Robotics all ne

ସ୍ୱାଧୀନ ଚାଲିବା "Dardical devices

dis Industrial automation IV) Aircraft
control.

Interaction Diagram

User

process *System Calls*

RTOS kernel

1

Schedulars Memory

Driver

Hard ware,

9 Process

Synchronization

System calls

1. forcecs pr creates process

2. wait

o

waits for another process

3. sem_wait

Locks semaphore

to

4 . sem post

unlocks semaphore.

5. pthread_mutex_lock

Locas shared resource.

6 . pthread_mutex_unlock c

g

Releases Shared resources

2014

These system calls prevents

race conditions

Synchronized execution.

and

Ensure

7

Char buffer!

TOO I

Ce

classity the

Centralized

the os and compare it with

Distributed operating
System...

A distributed inter connected

Advantages:

OS manages several. Computers as one

احمد

i) Resource sharing 1) load balancing

1) Fault tolerance

iv) scalability

performance

1 High performance comparison.

System.

Z

camb

Costines

Distributed as
centralized os

multiple computer
Single computer,

High reliability,

simon

Lower scalability

Better Scalability s

limited scalability

Faster resource shoot

Shooung

19

Local resources only

Fault tolerant

single point of

failure.

b) Architecture **ager bis 20**

Node I

arts plizer

bakdo

20

Networktops batudado

1

1

Node 2

Nodes

Node 4,

catosan s

1

Local

Local

Local

bernel
bernel

9) 1pc System calls

Pipecs forkcs

readcs

writes

msggete

msggroves **Shiget** Shmato demgeto

These calls enable

berrel. **Lopos** tarp

c

wordt tus (

20

votopis stiklus

communication between

wilde processes through
pipes, shared memory

L

who message queves, and
Semaphores.

२. ३. ४. ५.

thing alpriss

troot Flust

major functions of an

process Management ;) Creates
processes

ii) Schedules (priority)

Operating system

10:00 2pm.

notheballs

639

Ball e promis

iii) Terminates processes

iv) content

switching **alle** winst.

Memory Management

i) Allocates Rama

أو ملأه

mog och abala

noikos Ugg A

" ") Deallocates memory

Mij virtual memory to
part.

IN Paging

4) degmentation.

Device Management

I control 10 devices, `ond ii)`

Device driverspishion

11.) Buffering

iN) Spooking.

File Management.

`Aelord`

`I`

`ramsh`

`#`

`srep colves H`

`llo`

froment

1 organize tiles

d'iterolls poste 11)

Creates and delete tiles to

lii) controls file access.

D

Rollon

5. Security

Authentication do

solbord of

10

tiemporang B

Authorization

Access control.

6

Resource Allocation.

CPU Scheduling

duling

woord whoor) (9b8 (4

Memory allocation.

istoms? (

puntos bathos (vi

Device allocation.

Block diagram

Users

transpoint fromsty

Application programs

1

operating System

MAS kosa A

process Management Memory

Management File raanagement

Device Management. Security

Hardware.

۱۱ —

Louth in

25490 (1)

ان وسلامتها

System call Examples.

Memory

allocation

malloees

free mmapo

bok

09

no

Worche

┌

Open c

read C write

octles

closec

These

allocate

**Byston calls allow
program to
memory
dynamically and
communicate
with hardware
devices.**